

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Fifth Semester B.Tech (CBCS) Degree Examinations

December 2024

AM3201-1 – CLOUD COMPUTING

(For AM)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- In which computing approach do processes share memory and resources?
 - Distributed Computing
 - Parallel Computing
 - Cloud Computing
 - Local Computing
- Which of the following is a type of hardware architecture for parallel processing?
 - Master-Slave Architecture
 - Single-Instruction Multiple-Data (SIMD)
 - Server-Client Architecture
 - None of the above
- What is the main purpose of a Remote Procedure Call (RPC) in distributed computing?
 - It ensures data consistency
 - It allows a program to execute a procedure on a remote server
 - It monitors network traffic
 - It enhances CPU performance
- Which of the following is NOT an architectural style for distributed computing?
 - Client-server
 - Peer-to-peer
 - Monolithic
 - Microservices
- Which of the following cloud models provides hardware resources such as servers, storage, and networking?
 - Software as a Service (SaaS)
 - Platform as a Service (PaaS)
 - Infrastructure as a Service (IaaS)
 - Function as a Service (FaaS)
- Which type of cloud is managed by a third-party provider and accessible by the public?
 - Private Cloud
 - Public Cloud
 - Hybrid Cloud
 - Community Cloud
- What is the main advantage of hybrid clouds?
 - Complete control over data
 - Cost efficiency
 - Flexibility by combining public and private cloud benefits
 - Better data security than private clouds
- Which of the following is NOT a type of virtualization?
 - Compute Virtualization
 - Storage Virtualization
 - Desktop Virtualization
 - Network Virtualization
- Which of the following is a popular virtualization technology?
 - XEN
 - Apache
 - Hadoop
 - PostgreSQL
- Which is a common security concern in cloud computing?
 - Data Breaches
 - Lower Performance
 - Increased Hardware Cost
 - Slower Internet Speeds
- What does the C.I.A. triad in cloud security stand for?
 - Confidentiality, Integrity, Availability
 - Computing, Information, Accessibility
 - Compliance, Integration, Accountability
 - Cloud, Intelligence, Analysis

12. Which stage of risk management focuses on continuously monitoring security controls?
 A) Risk Identification B) Risk Assessment
 C) Risk Monitoring D) Risk Mitigation
13. Which organization provides guidelines on cloud security controls and risk management?
 A) NIST B) FDA
 C) ISO D) IEEE
14. What is a key best practice for cloud service consumers to ensure security?
 A) Rely solely on the cloud provider's security B) Implement encryption and access control
 C) Avoid using multi-factor authentication D) Store all data in a public cloud
15. What is the purpose of security monitoring in cloud computing?
 A) To track cloud performance B) To detect and respond to security threats
 C) To reduce cloud storage costs D) To automate cloud deployment
16. What does Monitoring as a Service (MaaS) offer?
 A) Automatic resource provisioning B) Security monitoring capabilities as a cloud service
 C) Improved network bandwidth D) Real-time data analytics
17. Which of the following is an example of a private cloud platform?
 A) AWS B) Google App Engine
 C) OpenStack D) Microsoft Azure
18. What is the main goal of risk management in cloud computing?
 A) To reduce costs associated with hardware maintenance B) To ensure the cloud provider meets its security obligations
 C) To identify, assess, and mitigate security risks D) To improve the speed of data transmission
19. Which of the following is a best practice for cloud service consumers?
 A) Avoid encrypting sensitive data B) Rely solely on the cloud provider's security controls
 C) Implement multi-factor authentication for data access D) Store all data in public clouds
20. What is the primary purpose of security monitoring in cloud environments?
 A) To track cloud performance B) To detect and respond to security threats
 C) To optimize network speed D) To reduce cloud storage costs

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) What is distributed system? Explain the layered view of a distributed system.	8	L*2	1	1
	b) Explain the following hardware architectures for parallel processing: a) Single-instruction, multiple-data (SIMD) systems b) Multiple-instruction, multiple-data (MIMD) systems	8	L2	1	1
2.	a) What are the main characteristics of a Platform-as-a-Service solution?	5	L2	1	1
	b) Explain the different models for message-based communication in distributed computing.	5	L2	1	1
	c) Describe the different levels of parallelism that can be obtained in a computing system.	6	L2	2	2
3.	a) What are the economic benefits of adopting cloud computing for businesses? Discuss factors such as cost savings, scalability, and flexibility.	8	L2	2	1
	b) Explain Service-Oriented Architecture (SOA) in the context of distributed computing. How does SOA contribute to the scalability and modularity of distributed systems, particularly in cloud environments?	8	L2	2	2

Unit – II

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|-------|---|---|----|---|---|
| 4. a) | What are the primary security concerns associated with cloud computing? How do these concerns vary across different cloud models? | 8 | L2 | 3 | 1 |
| b) | Discuss the relationship between virtualization and cloud computing. | 8 | L2 | 3 | 2 |
| 5. a) | What are the different types of virtualization other than execution virtualization? Briefly explain each of them. | 8 | L2 | 3 | 2 |
| b) | Explain the NIST definitions for security controls in cloud environments. | 8 | L2 | 4 | 1 |
| 6. a) | What are the limitations of security controls in cloud environments? How do evolving security threats challenge the effectiveness of traditional cloud security measures? | 8 | L2 | 4 | 1 |
| b) | Discuss the limits of security controls in cloud computing. | 8 | L2 | 4 | 2 |

Unit – III

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|-------|---|---|----|---|---|
| 7. a) | What is the purpose of security monitoring in cloud environments? How does security monitoring help detect and respond to threats in real-time? | 8 | L2 | 5 | 1 |
| b) | What are the key security monitoring features provided by AWS? How does AWS ensure the security of its public cloud environment for users? | 8 | L2 | 5 | 2 |
| 8. a) | How does Eucalyptus support security monitoring and management in private cloud deployments? | 8 | L2 | 5 | 1 |
| b) | What are the key security benefits and challenges associated with using AWS as a public cloud provider? | 8 | L2 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
